



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JUNE 2024**

Course: EC3130 (Introduction to Data
Communication and Networking)
Lecturer: Mr. SEOW Soon Loong

Time: 2:00pm – 5:00pm (3 hours)
Date: 20 September 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Explain the differences between the IEEE 802.11 standards: 802.11a, 802.11b, 802.11g, 802.11n, and 802.11ac. Focus on their frequency bands, data rates, and compatibility. (15 MARKS)

- b. Compare and contrast multimode and single-mode fibre-optic cables. Discuss their respective advantage and disadvantage. (10 MARKS)

2. a. You are tasked with designing a company network using Variable Length Subnet Masking (VLSM). The network requirements include four Local Area Networks (LANs) and two Wide Area Networks (WANs). The LANs will serve the following departments: (15 MARKS)
 - Administration (16 hosts)
 - Human Resources (6 hosts)
 - Marketing (30 hosts)
 - Operations Management (48 hosts)

Additionally, there will be two WANs, WLink1 and WLink2, to support daily operations. The company's network will use the Class C address 192.168.10.0/24.

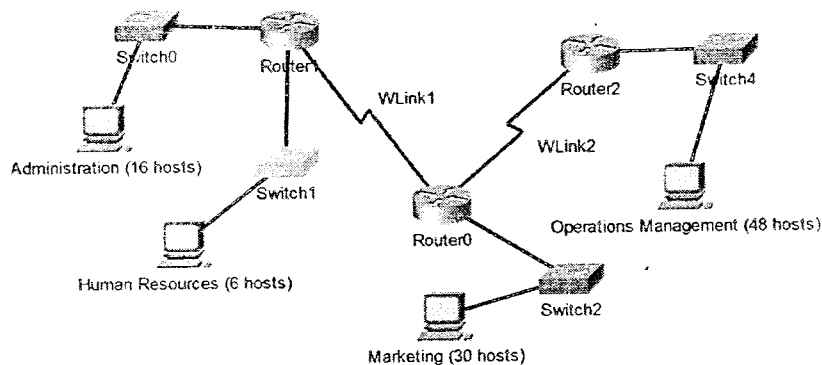


Figure 2.1: Network Requirements

Design the network addressing scheme using VLSM to meet the above requirements. Clearly indicate the subnet addresses, subnet masks, the range of IP addresses, and the broadcast address for each LAN and WAN using the format given in Table 2.2.

Subnet Name	Subnet address	First usable host address	Last usable host address	Broadcast address	Subnet Mask

Table 2.2

- b. How many usable host addresses are available for the **FOUR (4)** subnets created for each of the following LANs: Administration department, Human Resources, Marketing, Operations Management; and the WAN Links (WLink1 & WLink2)? **(6 MARKS)**

Subnet Name	# of usable host addresses
Administration department	
Human Resources	
Marketing	
Operations Management	
Wlink1	
Wlink2	

Table 2.3

- c. Provide a step-by-step example of shortening the following IPv6 address using the rules for IPv6 address compression:
- i. 2001:0db8:0:0042:0000:8a2e:0370:7334 **(2 MARKS)**
 - ii. fe80:0000:0000:0000:0202:b3ff:fe1e:8329 **(2 MARKS)**
3. a. Carrier-Sense Multiple Access with Collision Detection (CSMA/CD) is a network protocol used in Ethernet networks to manage access to a shared communication medium. Explain the main steps involved in the CSMA/CD process. **(18 MARKS)**
- b. Channelization is a multiple-access method used in wireless communications where the available bandwidth of a link is shared among different stations.
- i. Explain the principles of Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA) in the context of channelization. **(6 MARKS)**
 - ii. Discuss one advantage of CDMA. **(1 MARKS)**
4. a. Nonreturn to Zero-Level (NRZ-L) and Nonreturn to Zero Inverted (NRZI) are binary encoding schemes used in digital communications. Given the binary data stream: 0100110011, create TWO (2) diagrams to represent the signal for NRZ-L and NRZI. **(10 MARKS)**

- b. Huffman coding can be used in communication protocols to reduce the amount of data that needs to be transmitted. Consider the following set of characters and their corresponding frequencies: (15 MARKS)

- A: 10
- B: 15
- C: 12
- D: 3
- E: 4
- F: 13
- G: 1

Construct the Huffman Tree for the given characters. Draw the tree and indicate the Huffman codes for each character.

-END OF QUESTION PAPER-